



A Review of Causal Inference Methods for Estimating the Effects of Exposure Change when Incident Exposure Is Unobservable

Fangyu Liu^{1,2} · Emilie D. Duchesneau³ · Jennifer L. Lund⁴ · John W. Jackson^{1,2,5}

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Abstract

Purpose of Review Research questions on exposure change and health outcomes are both relevant to clinical and policy decision making for public health. Causal inference methods can help investigators answer questions about exposure change when the first or incident exposure is unobserved or not well defined. This review aims to help researchers conceive of helpful causal research questions about exposure change and understand various statistical methods for answering these questions to promote wider adoption of causal inference methods in research on exposure change outside the field of pharmacoepidemiology.

Recent Findings Epidemiologic studies examining exposure changes face challenges that can be addressed by causal inference methods, including target trial emulation. However, their application outside the field of pharmacoepidemiology is limited.

Summary In this review, we (a) illustrate considerations in defining an exposure change and defining the total and joint effects of an exposure change, (b) provide practical guidance on trial emulation design and data set-up for statistical analysis, (c) demonstrate four statistical methods that can estimate total and/or joint effects (structural conditional mean models, time-dependent matching, inverse probability weighting, and the parametric g-formula), and (d) compare the advantages and limitations of these statistical methods.

Keywords Target trial emulation · Exposure change · Sequential conditional mean model · Time-dependent matching · Inverse probability weighting · Parametric g-formula

Introduction

For many exposures of interest in epidemiologic studies, individuals can move back-and-forth between categories of exposure (e.g., exposed and unexposed) over the life course.

To date, the field has often focused on assessing how time-varying levels of exposures affect health outcomes. Less attention has been given to the effect of specific forms of change in exposure (i.e., what we call exposure change), which can, in some settings, be more relevant to clinical and policy decision making for some exposures. In addition, a definition of exposure change requires consideration of prior exposure levels and hence can mitigate bias introduced by mixing persons who are newly exposed with those who have been exposed for a long time.

Effects of exposure changes are clinically and policy-relevant questions, especially for complex exposures for which the first time an individual becomes exposed is unobservable or not well defined. For example, physical activity (PA) is an important protective factor for chronic disease and functional decline with aging. Less than 15% of older adults achieve the recommended 150 min per week of moderate-to-vigorous PA, which may contribute to adverse health outcomes in this population [1]. Research on the effect of increasing PA, an exposure change, among

✉ Fangyu Liu
fliu28@jhu.edu

¹ Department of Epidemiology, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA

² Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA

³ Department of Epidemiology and Prevention, Division of Public Health Sciences, Wake Forest School of Medicine, Winston-Salem, NC, USA

⁴ Department of Epidemiology, Gillings School of Global Public Health, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

⁵ Center for Drug Safety and Effectiveness, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD, USA

older adults may help inform guidelines on physical activity. For example, research questions on the effects of the extent or patterns of PA change (e.g., staged increase over a longer vs shorter period of time, or PA performed in longer vs shorter bouts) among different subpopulations (e.g., extremely sedentary individuals or individuals living with chronic conditions) can also inform how public health efforts can be targeted.

Studying exposure change also mitigates the potential for bias from the mixing of prevalent and incident exposures. This bias occurs more frequently when studying the effect of an exposure level when the first visit with an exposure assessment is used as the time zero for baseline exposure. This approach includes individuals who have been exposed previously (prevalent exposure), individuals who are newly exposed (incident exposure), and individuals who are unexposed. Because individuals who have been exposed to a harmful exposure for some time without experiencing the outcome may be healthier than individuals who are newly exposed or unexposed, a bias sometimes called the “healthy adherer bias” [2••] can be introduced. The definition of an exposure change requires researchers to define from what prior exposure level a change is made and when such a change is made (e.g., after an eligibility event). Therefore, it helps mitigate this bias by providing a better-defined baseline to evaluate eligibility, assess covariates, and to start follow-up.

The design of studying an exposure change after an eligibility event has been introduced in the pharmacoepidemiology, termed the “treatment decision design” [3••]. This design closely follows causal inference methods, including the target trial emulation framework [4••], and may also be useful for studying non-pharmacologic exposure changes, especially for complex exposures for which the first time an individual becomes exposed (incident exposure) is unobservable or not well defined. The target trial emulation framework requires specification of the target trial protocols including eligibility criteria, intervention strategies, causal effect of interest, and analytic plan as if such a trial is feasible and ethical, and description of how the observational study design can mimic or emulate the protocol [4••]. Causal inference methods help avoid common pitfalls when studying exposure changes, such as confounding bias where individuals who increases a protective exposure (e.g., PA) may be healthier than individuals who does not initiate such an increase (sometimes called the “healthy initiator bias”) [2••, 5]. This bias can be addressed by careful selection of confounders in the causal inference framework.

Despite the relevance of research questions on exposure change and the availability of causal inference methods to study these questions, few epidemiologic studies outside the field of pharmacoepidemiology have utilized these methods. There are several barriers to wider adoption. First, there is a

lack of guidance on how to define exposure changes, conceptualize the relevant effects of interests, set up the data, and apply the target trial emulation framework in non-pharmacological contexts. Second, there are multiple causal inference methods for estimating different causal effects of interest. Their application to the investigation of exposure changes, and their strengths and limitations in this setting are not well described for applied audiences. In this review, we discuss causal inference methods for studying non-pharmacologic exposure changes using a simplified motivating example of the effect of PA change after a hypertension diagnosis on the 5-year risk of myocardial infarction (MI). This review provides a helpful guide for researchers who are interested in using causal inference methods to study exposure change, particularly when there is not a clearly defined and observed incident exposure.

Defining Exposure Change

In our motivating example, we define PA change as increasing PA to ≥ 150 min/week (guideline level) immediately after hypertension diagnosis among individuals who had PA < 150 min/week for at least a year before the diagnosis. There are a few elements in this definition that can be altered based on scientific interest and relevance. The first element is *the eligibility for exposure change*. In our definition, a hypertension diagnosis is the eligibility event and only individuals who did not meet the PA guideline before the diagnosis are eligible for the defined exposure change and are included in the analysis. Note that our definition of change includes the phrase “at least a year before diagnosis” which specifies the duration of the qualifying condition that persons are to change from. The second element is *the threshold to meet the exposure change*. Here, we specify a uniform threshold (i.e., ≥ 150 min/week) for all eligible individuals (static strategy), but the threshold could vary dependent on values of some covariates (dynamic strategy) [6]. For example, we could specify a higher threshold (e.g., ≥ 170 min/week) for participants with systolic blood pressures ≥ 140 mmHg but retain the original threshold (i.e., ≥ 150 min/week) for participants with systolic blood pressures between 130 and 140 mmHg. The last element in defining an exposure change is *the time frame for change*. Instead of an immediate change, a grace period may be given that allows time for individuals to make the required change (e.g., within 2 years of diagnosis) [7, 8]. We give more detailed examples of these possible variations in Table 1.

Data availability can influence these elements of an exposure change definition. Is the PA assessment valid and feasible to determine whether an individual met the threshold for change? If the PA questionnaire only collects the frequency of a few types of activity, an exposure change defined as a

Table 1 Exposure change defined in this review and the possible variations

Elements of exposure change definition	In the review	Example(s) of possible variations
Eligibility for exposure change	Only individuals who had < 150 min/week of physical activity prior to hypertension diagnosis for at least a year are eligible	Everyone is eligible for the exposure change, regardless of their physical activity level prior to hypertension diagnosis
Threshold for exposure change	Increase physical activity to recommended level, i.e., ≥ 150 min/week	Increase physical activity by a percentage (e.g., increase by 20% from current level)
Dynamic exposure change	Static exposure strategy	Threshold for increase in physical activity depends on systolic blood pressure (e.g., SBP ≥ 140 mmHg should achieve ≥ 170 min/week whereas those with 130 to 140 mmHg should achieve ≥ 150 min/week)
Grace period for exposure change	No grace period (exposure change must occur immediately after the hypertension diagnosis)	Increase in physical activity occurs within 2 years of hypertension diagnosis

change in frequency may be more relevant than a change in total activity duration. Does the data source have sufficient historical data to identify the eligibility event? If the data source does not collect medical records of the participants before their enrolment of the study, new diagnosis of hypertension may not be reliably identified. In such cases, using a new hypertension diagnosis as the eligible event may not be feasible. On the other hand, an eligible event of recent failure to control hypertension may require a shorter history and be more feasible with the data source. Balance between scientific interest and data availability should be considered when defining exposure change.

For simplicity, we limit our example to a static exposure change without a grace period. Several papers have specifically discussed how to estimate the effects of more complex scenarios with a dynamic strategy or a grace period [6, 7, 9, 10•]. Our discussions are also relevant to methods that estimate the effect of increasing the probability of achieving the guideline of 150 min/week [11], though we do not review them here.

Relevant Effects of Interest for Exposure Change

The total effect and joint effects of time-varying exposures are two commonly estimated effects that are relevant to clinical decisions and policy making and are the focus of this review. Other types of effects such as the “controlled direct effect” [12] or, equivalently, “long-term direct effect” [13••] of time-varying exposures are summarized elsewhere.

We refer to the *total effect* of an exposure change as the comparison of outcomes for individuals who initiate the defined exposure change at baseline to those for individuals who do not initiate the exposure change, regardless of the subsequent exposure status between baseline and the end of follow-up (Fig. 1). The total effect is akin to the intention-to-treat effect in a randomized control trial. In our example,

the total effect compares the outcomes between individuals who increased PA to ≥ 150 min/week immediately after the hypertension diagnosis (baseline) and those who did not (Table 2). An individual who increased PA immediately after the hypertension diagnosis to > 150 min/week may fall back to < 150 min/week after a few years, and an individual who did not immediately increase PA may eventually increase PA. The subsequent non-adherence to the exposure changes is captured by the total effect.

We refer to the *joint effect* of an exposure change as the comparison of outcomes under a sustained exposure change at baseline and throughout follow-up (Fig. 1). The joint effect is akin to the per-protocol effect in a randomized control trial. In our example, the joint effect compares the outcome between individuals who increased PA to ≥ 150 min/week immediately after the hypertension diagnosis and maintained this level during the follow-up and those who never initiated the increase (Table 2). It is worth noting that a joint effect, generally, involves the causal effect of intervening on multiple distinct exposures (e.g., physical activity and diet) which often arises in the setting of causal interaction [14]. Here, when we reference the joint effect, we are interested a special case, the causal effect on intervening on a single type of exposure that occurs at multiple times (e.g., physical activity at baseline and at subsequent visits) which arises in the setting of studying effects of cumulative or sustained exposures.

Specifying the effect of interest helps guide the selection of suitable methods for effect estimation. When the total effect is of interest, what is called “exposure-confounder feedback” [6] is not of concern. In our example, exposure-confounder feedback describes a situation where a covariate (e.g., body mass index) measured at one timepoint affects both subsequent achievement of ≥ 150 min/week of PA and MI and is also affected by achievement of ≥ 150 min/week of PA at an earlier timepoint. When the total effect is of interest, only achievement of ≥ 150 min/week of PA or the lack of it *at baseline* (i.e., immediately after the hypertension

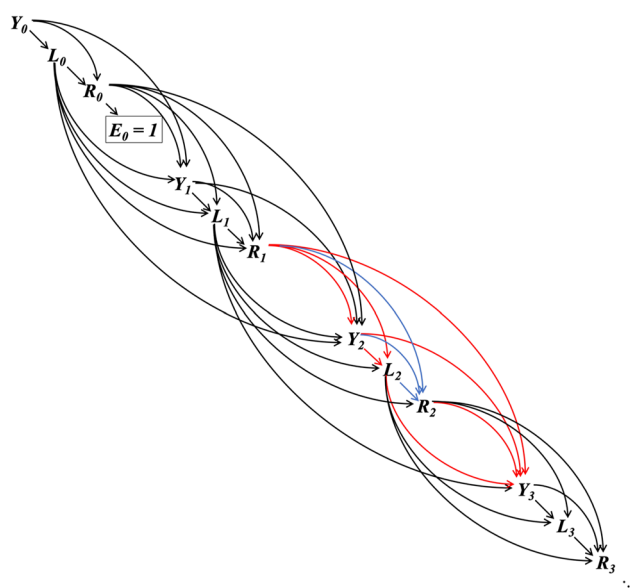


Fig. 1 Causal Diagram for the illustration example. Notes. R_t denotes whether physical activity ≥ 150 min/week was achieved at each visit $t = 1, 2, 3, \dots$. Since the sample consists of individuals who had < 150 min/week of physical activity before hypertension diagnosis ($E_0 = 1$), R_t denotes exposure changes. L_t denotes covariates including body mass index, smoking status, drinking status, and cholesterol level. L_t also include time-fixed covariates—age, sex, race, and a summary variable of physical activity history prior to the diagnosis. Y_t denotes the outcome, myocardial infarction. We assume the temporal ordering of variables at each visit as (Y_t, L_t, R_t, E_t) . Exchangeability between R_t and Y_{t+1} at each time t is assumed to hold after conditioning on L_{t-1}, R_{t-1}, Y_t , and L_t . The red and blue paths are both part of the total effect of R_1 on Y_3 . For total effect, R_1 alone is the exposure change of interest, and R_2 is not an exposure. Therefore, paths such as $R_1 \rightarrow L_2 \rightarrow R_2$ are not “exposure-confounder” feedback (i.e., there is no “exposure-confounder” feedback). The red paths are part of the joint effect of R_1 and R_2 on Y_3 . The joint effect also includes any casual interactions present between R_1 and R_2 which are consistent with the diagram. For the joint effect, R_1 and R_2 are both exposures under the change of interest. Therefore, an “exposure-confounder feedback is present as L_2 , a confounder of R_2 and Y_3 , is affected by R_1

diagnosis) defines whether an individual initiates the exposure change. Though we have subsequent measurements of PA *after baseline* and can calculate whether these PA measurements represent an increase from pre-diagnosis measurement, these subsequent PA changes are not part of the exposure change under the total effect. Likewise, pre-baseline PA measurements are not part of the exposure change of interest, since they occur prior to the eligibility event (although, pre-baseline PA measures may serve as important confounders in the statistical analysis). Under this framing of the total effect, there is only a single timepoint of PA change per person, and hence, there is no possible feedback loop (Fig. 1) for the total effect.

Though exceptions exist, when the joint effect is of interest, exposure-confounder feedback is often present. In our example, achievements of ≥ 150 min/week of PA at baseline and *all follow-up* visits are considered instances of exposure change, and exposure-confounder feedback between PA and covariates (e.g., body mass index) is possible (Fig. 1B). If the feedback is present, standard regression models are no longer appropriate for estimation of the joint effect [6]. Instead, marginal structural models (MSM) via inverse probability weighting (IPW) or the parametric g-formula method are required for valid estimation.

Which effect is of interest should be driven by scientific interest and knowledge, but data feasibility can also influence which effect can be casually identified by evaluating whether conditions of exchangeability (i.e., no unmeasured confounding), positivity, and consistency are met. Please see Appendices A and B for a more detailed discussion. Note that, when the exposure and outcome is repeatedly measured, measures of pre-baseline outcomes and exposures are often recommended as confounders to satisfy exchangeability [15]. We follow this recommendation through our

Table 2 Research questions considered in this review

Type of effect	Research question	Applicable analytical methods
Total effect	Q1. Among individuals who had physical activity < 150 min/week for one year before hypertension diagnosis, does increasing physical activity to ≥ 150 min/week immediately after the diagnosis reduce the 5-year risk of MI, compared to not increasing physical activity immediately after diagnosis?	Sequential conditional mean model Time-dependent matching Marginal structural model via inverse probability weighting Parametric g-formula
Joint effect	Q2. Among individuals who had physical activity < 150 min/week for one year before hypertension diagnosis, does increasing physical activity to ≥ 150 min/week immediately after the diagnosis and maintaining the increased physical activity level reduce the 5-year risk of MI, compared to maintaining the physical activity level at diagnosis?	Marginal structural model via inverse probability weighting Parametric g-formula

eligibility criteria, i.e., by excluding individuals with a history of MI at baseline and those with PA ≥ 150 min during the year prior to baseline.

Target Trial Emulation, Data Source, Data Set-up, and Timescale

To answer the research questions in Table 2, we can emulate hypothetical target trials. The key protocol components of the hypothetical trial and observational study are summarized in Table 3 using data from a hypothetical observational cohort of seven individuals who are followed annually in calendar time (Fig. 2A). To exclude individuals based on target trial eligibility criteria, we use the temporal ordering of the variables measured at the same visit of the observational cohort. This is ordering is usually assumed as it is challenging to tell if a variable precedes the other when they are measured at the same visit. However, the instruments used for variable measurement may sometimes inform the plausible temporal ordering. For example, suppose hypertension is diagnosed using the systolic and diastolic blood pressures measured *at the visit* and PA is measured by self-report of the behavior in *the past year before the visit*. It would likely be unreasonable to assume that an incident hypertension diagnosis made at the time of the visit precedes the PA measured at the same visit. (Although, certainly, if a hypertension diagnosis were made at the previous visit, it could influence PA.) In this review, we assume the temporal ordering of variables measured at the same visit is MI, covariates, PA, and hypertension. Based on this temporal ordering, we exclude ID 6 from the analysis because their MI event occurred before the hypertension diagnosis and ID 7 because MI occurred before initiating/forgoing the exposure change.

The target trial would randomize individuals when they were diagnosed with hypertension and follow them thereafter. Therefore, the appropriate analytical timescale for the emulated trial is “time since the eligibility event” (see Appendix A for a discussion on other appropriate timescales for other types of research questions). Since the timescale of the data source is calendar time, the source data need to be re-aligned by hypertension diagnosis (Fig. 2B). Note here that the baseline in the emulated trial is one visit after hypertension diagnosis because of the temporal ordering we assume. Calendar time is the most common timescale for cohort studies, but other timescales exist, e.g., time since an event or age. If the source cohort enrolled participants when they were diagnosed with hypertension, re-alignment of the timescale may not be necessary.

Statistical Methods for Estimating Total and Joint Effects of Exposure Change

We highlight four methods that can estimate the total effect of an exposure change: (1) the sequential conditional mean model (SCMM) [13••], (2) time-dependent matching [16••], (3) the marginal structural model via inverse probability weighting (MSM via IPW) [17•], and (4) the parametric g-formula [18]. The MSM via IPW and the parametric g-formula can estimate the joint effect of an exposure change in the presence of “exposure-confounder feedback” (Table 2). We discuss the procedures of each method, assumptions to be considered, and their specific advantages and disadvantages. We illustrate the step-by-step implementation of each method in Appendix C.

Causal Structure and Causal Effect Measures

We assume a causal structure as depicted in Fig. 1 for our research questions. Exchangeability between PA change and MI at the next visit is established after conditioning on covariates measured at the current and prior visits, PA change at the prior visit, and MI measured at the current visit (the last one being enforced by restricting to those without history of MI). The covariates include time-varying covariates—body mass index, smoking status, drinking status, and cholesterol level, and time-fixed covariates—age, sex, race, and calendar year at baseline. For didactic purposes, we have simplified the exposure to a binary indicator of ≥ 150 min/week and have ignored that PA duration is a continuous measure and that PA activity involves different forms of PA. In practice, these considerations can influence which variables are needed to satisfy conditional exchangeability [19].

The outcome of interest, MI, is a time-to-event outcome. One common effect measure of a time-to-event outcome is the hazard ratio, most frequently estimated using a Cox proportional hazard model. However, the hazard ratio has a challenging causal interpretation because it is a conditional quantity (conditional on surviving the previous visit) and hence subject to selection bias [20, 21]. The alternative effect measures use the cumulative incidence or risk of MI as we stated in the research questions (Table 2)—risk difference, risk ratio, and odds ratio. Each of these effect measures is a valid causal estimand. For simplicity, we focus on the risk ratio for both the total effect and joint effect estimation and make our analytical sample (Fig. 2B) free of right censoring due to competing risk and free of loss to follow-up. We discuss relevant statistical methods when right censoring is present and provide suggested references for estimation under competing risks, in Appendix D.

Table 3 Target trial specification and emulation using observational data for the motivating example of the causal effect of physical activity change on the risk of myocardial infarction

	Hypothetical target trial	Observational study emulating the target trial
Eligibility criteria	1. Newly diagnosed with hypertension 2. < 150 min/week of physical activity during the year prior to the hypertension diagnosis 3. No history of MI 4. No contraindications to physical activity	1. Newly diagnosed with hypertension 2. < 150 min/week of physical activity during the year prior to the hypertension diagnosis 3. No report of MI prior to or at the visit when hypertension was diagnosed 4. No mobility disability measured by gait speed
Treatment strategy	Randomized to either (1) increase physical activity to ≥ 150 min/week or (2) maintain current physical activity level (< 150 min/week) immediately after hypertension diagnosis	Observed treatment, either: (1) increase physical activity to ≥ 150 min/week or (2) maintain pre-diagnosis physical activity level (< 150 min/week) at the first visit following the hypertension diagnosis
Causal effect of interest	1. Total effect: the effect of being assigned to increased physical activity strategy at baseline compared to being assigned no increase from pre-diagnosis physical activity level at baseline 2. Joint effect: the effect of increasing physical activity at baseline and adhering to the increased physical activity throughout the follow-up compared to maintaining the pre-diagnosis physical activity at baseline and throughout the follow-up	1. Total effect: the effect of increasing physical activity at baseline compared to no increase from pre-diagnosis physical activity level at baseline 2. Joint effect: the effect of increasing physical activity at baseline and adhering to the increased physical activity throughout the follow-up compared to maintaining the pre-diagnosis physical activity at baseline and throughout the follow-up
Analytic plan	1. Total effect: intention to treat analysis 2. Joint effect: comparison of MI risk between the two groups with adjustment for pre- and post-baseline prognostic factors associated with adherence to the intervention and loss to follow-up including death (per protocol analysis)	1. Total effect: comparison of 5-year MI risk between the two groups with adjustment for pre-baseline prognostic factors associated with physical activity increase and loss to follow-up 2. Joint effect: comparison of MI risk between the two groups with adjustment for pre- and post-baseline prognostic factors associated with adherence to the intervention and loss to follow-up

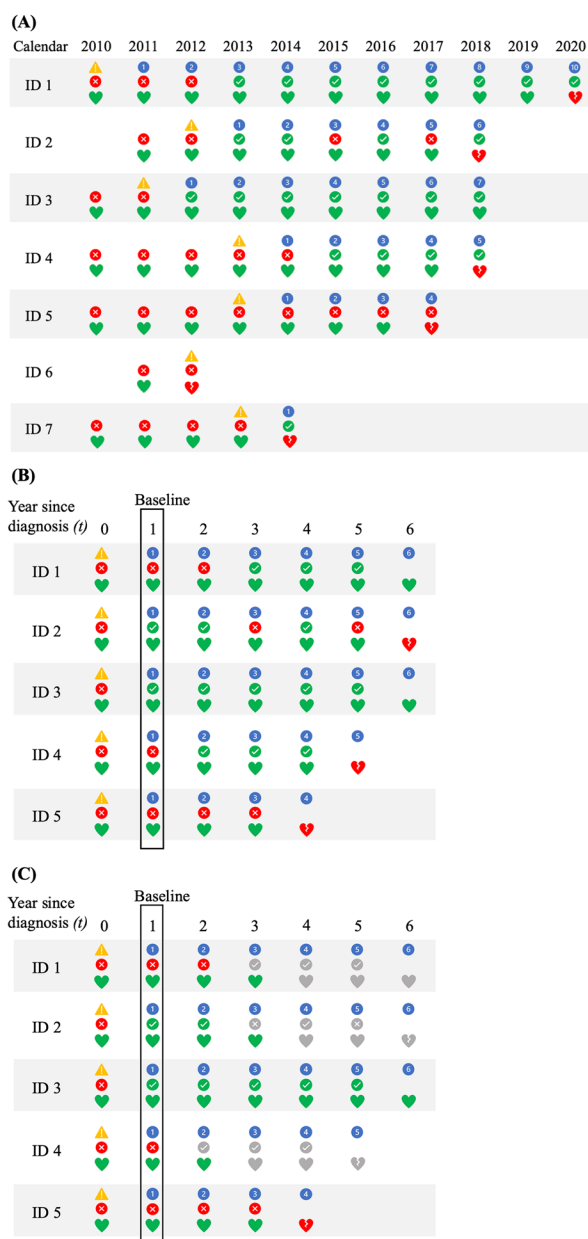


Fig. 2 Data structure in the observational cohort (A), in the analytical sample (B), and after censoring participants due to non-adherence to the PA change that they adopt at baseline (C). Notes. The yellow caution symbols denote the visit when hypertension is first diagnosed using systolic and diastolic blood pressures measured at the study visit. The blue circles with numbers following the yellow caution symbol mark the number of visits since hypertension diagnosis. The red circles with a cross sign denote self-report of having < 150 min/week of physical activity in the past year, whereas the green circles with a check denote having ≥ 150 min/week of physical activity in the past year. The green whole hearts in the third row denote no myocardial infarction (MI) before the current visit using medical records, whereas the red broken hearts denote MI at or before the current visit. Individuals are no longer followed after they have a first MI. The black box in (B) and (C) denotes the baseline for analysis. It is the first visit after hypertension diagnosis due to the temporal ordering of variables we assumed. (B) was truncated at Year 6 after hypertension diagnosis because we are interested in the 5-year risk of MI. Censored visits due to non-adherence at the PA change are denoted by the gray color icons in (C)

Sequential Conditional Mean Model

The SCMM estimates total effects. Because estimation of total effects is not challenged by the exposure change-confounder feedback, standard regression models can provide a valid causal estimate with correct model specification. Keogh and colleagues introduced the SCMM as pooling a series of standard regression models, each for an exposure initiated at a different timepoint along the analytical timescale [13••]. Since we defined that individuals were only eligible for exposure change at the next visit after hypertension diagnosis, there is no pooling of different timepoints, and the SCMM simplifies to a single standard regression model of the outcome, e.g., a modified Poisson regression model with robust variance for our outcome, with the indicator of PA change at baseline and all the covariates that establish exchangeability at baseline. The standard error from the model can be used to estimate the confidence interval of the causal effect.

Time-Dependent Matching

Time dependent matching is another method for estimating the total effect that pools exposure changes initiated at different timepoints [16••]. Though a change in PA (as we defined it) does not vary based on the time since the eligibility event, it can vary on other timescales, e.g., age, that may be associated with the exposure and outcome. In the SCMM, these other timescales are controlled as covariates in the model. In contrast, time-dependent matching can use them as the timescale for matching and non-parametrically controls for them. Using age as a timescale for matching requires the individuals to first be aligned by their age (Fig. 3A). At each age, we match individuals who had PA change to those who did not based on the age alone, age and a subset of categorical covariates, or a propensity score using all other covariates to establish exchangeability [16••].

After matching, the matched individuals are aligned by the eligibility event for analysis (Fig. 3B). When matching is based on age or a subset of categorical covariates (i.e., exact matching is possible for these variables), the total effect of PA change can be estimated using a modified Poisson regression model with robust variance with MI by the end of 5 years as the dependent variable and the PA change at baseline as the independent variable. To avoid bias, the model should also include the matching variables and any other covariates that are needed to control for confounding but are not used for matching [22]. If propensity score matching is performed, the modified Poisson regression model can include PA change at baseline alone (i.e., a marginal structural model). The variance estimate from the model output may not be valid if the observations were treated as

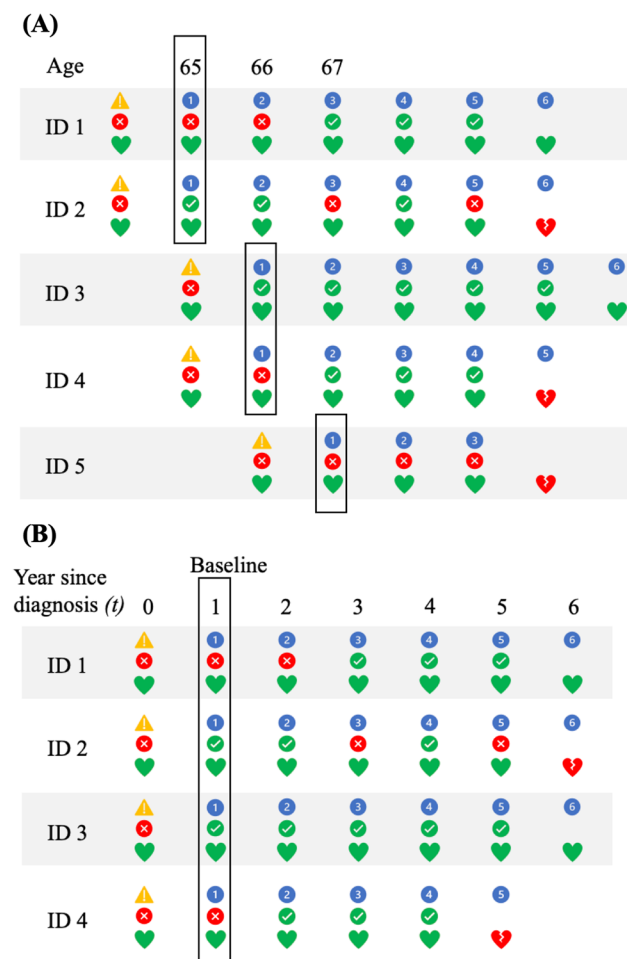


Fig. 3 Data alignment by age for estimating propensity score for matching (A) and by eligibility event for analysis (B). Notes. ID 5 is excluded in (B) because there was no match to them at age 67 years

independent. The bootstrap can be used to obtain the confidence interval of the effect estimate [23].

Marginal Structural Model via Inverse Probability Weighting

The MSM via IPW method uses weights to create a pseudo-sample in which covariates are balanced across PA change groups, as in a randomized control trial, and causal estimates can be obtained [13•, 24•]. When the total effect is of interest, the pseudo-sample is obtained by weighting the sample using inverse probability of treatment weights (IPTW) at baseline. IPTW are the inverse of the conditional probability of experiencing the exposure change individuals actually experienced (e.g., increasing PA or not). The weights can be estimated using a logistic model with PA change at baseline as the dependent variable, and all the covariates at baseline needed to establish exchangeability as the independent variables. If the covariates at baseline

are sufficiently balanced (across levels of baseline exposure change status) in the pseudo-sample [25], a marginal structural model using a weighted modified Poisson regression can be used to estimate the total effect of PA change on risk of MI. The bootstrap has been recommended to obtain the confidence interval in IPW method to account for the fact that the propensity scores are not known but estimated from the data [26].

When the joint effect is of interest, one approach to estimate the effect using MSM via IPW is implemented by artificially censoring visits when individuals no longer adhere to the PA change they adopted at the baseline (increase or no increase) during the follow-up [9]. For example, in Fig. 2C, ID 2 was censored at the third visit after hypertension diagnosis and thereafter because they failed to maintain the increased PA. Similarly, ID 1 was censored at the third visit after hypertension diagnosis and thereafter because they increased PA to ≥ 150 min/week and no longer met the definition for no PA increase. The motivation behind such censoring is to help reduce modeling assumptions (i.e., interaction terms with prior exposure are left unspecified as they are implicit). The MSM via IPW method can also be implemented without artificial censoring [27].

Then, using the artificial censoring indicators, we create inverse probability of artificial censoring weights to balance the covariates between uncensored and censored individuals at each visit [28]. Since the artificial censoring indicator is a one-to-one transformation of the indicator of exposure change (see Appendix A for details), these weights are equivalent to cumulative IPTW (i.e., the cumulative product of treatment weights at each visit) typically used to control for confounding, but in this case, the weights are used to address selection-bias induced by artificially censoring persons when they become non-adherent [9]. These weights are distinct from weights for censoring due to loss to follow-up which are discussed in Appendix D. These time-varying weights are estimated at each visit during follow-up based on the time-varying covariates at and prior to the visit using pooled logistic regression, which appropriately handles the exposure-confounder feedback. After weighting, the individuals who increased PA at baseline and maintained the increase through 5 years of follow-up create a pseudo-sample that represents everyone in the original sample if they had increased and maintained the increase of PA. Similarly, individuals who never increased PA throughout the 5-year follow-up create a pseudo-sample that represents everyone in the original sample if they had never increased PA. A time-varying form of covariate balance and weight distribution can be checked across levels of artificial censoring among those who initiated the PA change and among those that did not [24•, 28].

The drawback of the artificial censoring is that no right censoring is no longer true, and the estimation of the risk

ratio requires an adaptation of a Kaplan–Meier estimator or a cumulative incidence estimator, which are equivalent in the absence of competing risks [29]. To do so, we estimate the hazard of MI at each visit using a pooled logistic model weighted by the cumulative IPTW up to and including each visit. With the estimated hazard, we can estimate the 5-year risk of MI using a Kaplan–Meier or cumulative incidence estimator (i) when everyone increased PA to ≥ 150 min/week at baseline and maintained ≥ 150 min until 5 years after baseline and (ii) when everyone never increased PA to ≥ 150 min/week during the 5 years. The ratio between the mean of the two sets of risk is an estimate of the joint effect, and the confidence interval can be obtained via bootstrapping.

Parametric g-formula

The three methods described above estimate the causal effect of an exposure by comparing individuals who followed the strategy of exposure changes and those who did not, and properly accounting for non-adherence given the exchangeability assumption. The parametric g-formula method, on the other hand, predicts the outcomes by assigning *everyone* or *no one* to follow the strategy of exposure change under perfect adherence and then estimates the causal effect. The prediction of the outcomes is based on a conditional mean model of the outcome (e.g., the SCMM model described above). The prediction of the outcomes and subsequent expectation is equivalent to standardizing the conditional mean over the confounder distributions in the entire sample. The ratio between the two predicted/standardized means is a causal estimate of the effect of exposure change. For the total effect in our example, we compare the predicted risk of MI under a scenario in which *everyone* increased PA to ≥ 150 min/week at baseline versus the predicted risk when no one increased PA at baseline. For the joint effect, we compare the predicted risk of MI under a scenario in which *everyone* increased and maintained PA ≥ 150 min/week at baseline and throughout follow-up versus the predicted risk when *no one* increasing PA at baseline and throughout follow-up.

There are two main approaches to implement parametric g-formula: one using Monte Carlo simulation that requires models for the distribution of the covariates (noniterative conditional expectation or NICE) [18, 30•, 31•, 32] and the other using the empirical distribution of the covariates (iterative conditional expectation or ICE in the context of the joint effect) [10•, 33•]. When the total effect is of interest, the ICE approach involves estimating the conditional means using a SCMM model and predicting two outcomes from the model: (i) under the scenario that everyone increased their PA, by fixing the PA change indicator in the SCMM to 1 for everyone, and (ii) under the scenario that no one

increased their PA, by fixing the PA change indicator in the same SCMM to 0 for everyone. The observed covariate values are used for both predictions [33•]. The total effect is then estimated by the difference in means of the two sets of the predicted values, and the 95% confidence interval is obtained through bootstrapping. Alternatively, Monte Carlo simulation approach can also be applied [31•].

When the joint effect is of interest, both NICE and ICE approaches start by estimating the hazard of the outcome at each visit and can do so in the presence of the “artificial censoring” due to non-adherence. At each visit, both approaches control only covariates measured at and prior to that visit, hence appropriately handling the exposure-confounder feedback for the joint effect. The estimated hazards are used to calculate the 5-year risk of MI using a Kaplan–Meier or cumulative incidence estimator by sequentially averaging the hazards over the covariate history [29] for (i) everyone increased and maintained PA ≥ 150 min/week at baseline and throughout follow-up and (ii) everyone never increased PA at baseline and throughout the follow-up. The estimate of the joint effect is obtained by taking the ratio in the risks between the two groups, and confidence intervals are obtained via bootstrapping. Software implementations for NICE are available in R [34], Stata [35], and SAS [36], and an implementation for ICE is available in R [37].

Comparisons of the Statistical Methods

Though these four methods can unbiasedly estimate the total and/or joint effect of an exposure change, they vary in the estimands they target (Table 4). The estimand that SCMM provides is a conditional effect that is specific to covariate levels. In contrast, the parametric g-formula provides marginal effects that are typically averaged over the covariate distributions. Time-dependent matching and MSM via IPW can provide either conditional or marginal effect estimates. Under effect heterogeneity that is correctly modeled, conditional effects have a covariate-specific interpretation and are more relevant to research questions related to individual response (e.g., the reduction in MI risk given the covariate profile of an individual) [38]. Marginal effects, on the other hand, have a population-level interpretation and are more relevant to questions about policy (e.g., the reduction in MI risk in a population where the covariate distribution is the same as the study sample). In the presence of effect heterogeneity, a marginal effect is an average effect and depends on the distribution of the covariates. Without effect heterogeneity, estimates for conditional effects and marginal effects will be the same for mean differences (continuous outcome) and risk ratios or differences (binary or time to event outcomes) measures but will not be the same for odds ratios or hazard ratios due to non-collapsibility [39].

Table 4 Comparisons of the sequential conditional mean model (SCMM), time-dependent matching, marginal structural model via inverse probability weighting (MSM via IPW), and parametric g-formula

	SCMM	Time-dependent matching	MSM via IPW	Parametric g-formular	
				NICE	ICE
Estimand	Conditional effect ATE	Marginal or conditional effect ATT	Marginal or conditional effect ATE	Marginal effect ATE	
Assumptions for unbiased estimate	No misspecification of the SCMM	(1) Matching on time since hypertension diagnosis and a subset of categorical variables; no model misspecification of covariates included in the outcome model (2) Matching on propensity score; no model misspecification of propensity score model or prognostic score model	(1) Model for exposure is correctly specified (2) MSM model is correctly specified if it is unsaturated	Outcome models and covariates models are correctly specified	Outcome models are correctly specified
Advantages	(1) Easy to implement for all types of exposures (2) More efficient than MSM via IPW	(1) Specification of effect modification in MSM is optional (2) Nonparametric controlling of time and categorical covariates if they are the only matching criteria	(1) Specification of effect modification in MSM is optional (2) Less prone to model misspecification than NICE approach of parametric g-formula (3) No g-null bias	(1) Easy to implement for all types of exposures though the interpretation can be challenging (2) More efficient than MSM via IPW	
Disadvantage	(1) Effect modification must be correctly specified (2) Only estimate conditional effect	(1) Hard to implement for continuous exposure (2) Good match may not be available (3) Restriction to observations with an available match may affect generalizability	(1) Hard to implement for continuous exposure (2) Less efficient than other methods	(1) Only estimate marginal effect (2) G-null bias (3) Computationally intensive (4) More prone to model misspecification	

ATE average treatment effect, ATT average treatment effect among the treated, NICE noniterative conditional expectation, ICE iterative conditional expectation

Time-dependent matching typically is used to attempt to estimate an average treatment effect among the treated (ATT), provided there is sufficient covariate overlap to permit estimating the ATT, because we matched individuals that did not increase PA to those who increased PA (the treated). The matched sample represents only a subpopulation that has the same distribution of characteristics as those who were treated, which may differ from the distribution of characteristics in the entire population. In comparison, the other methods estimate an average treatment effect (ATE), which measures the extent that the risk of MI is reduced if everyone in the population increased their PA compared to everyone remained inactive.

Though meeting the assumptions of exchangeability, positivity, and consistency makes the estimates from these methods causal, whether the estimate is unbiased depends on the correct specification of the parametric models used in these methods (see Table 4 for details). There are doubly robust approaches to mitigate the potential for bias caused by model misspecification. Adjusting propensity scores in SCMM model [13••, 40] and fitting a conditional model with covariates instead of an MSM after time-dependent propensity score matching [16••] and procedures that combines IPTW and the ICE approach of the parametric g-formula [40–44] are some doubly robust methods.

Each method has strengths and limitations (Table 4). SCMM is easy to implement with regular regression models that epidemiologists are familiar with and can easily accommodate continuous exposures. Estimates from the SCMM are also more efficient (i.e., have smaller standard error) than estimates from the MSM via IPW [13••]. However, failure to correctly account for effect heterogeneity within the SCMM can result in bias. In contrast, when MSM is used with time-dependent propensity score matching or IPW, the estimate is still unbiased when effect heterogeneity is not specified in the outcome model, provided that the model for exposure change (i.e., the propensity score model) is correctly specified [13••].

The main advantage of the time-dependent matching method is controlling the confounding effect of the timescale used for matching nonparametrically, which avoids misspecification in a parametric model. However, the restriction of the study sample to observations with an available match may affect generalizability. Accommodation of continuous exposures is more challenging (but not impossible) [45].

The MSM via IPW method can be less prone to model misspecification than the NICE approach of the parametric g-formula method, as only the exposure model is needed. Moreover, MSM is not subject to g-null bias from which the parametric g-formula method suffers when there is exposure-confounder feedback [46••]. G-null bias means that the parametric g-formula method will reject the hypothesis that the exposure (at any time) does not have an effect on the

outcome even if this hypothesis is true [47]. Recent work has shown that it simply represents the guarantee of some bias due to model-misspecification, which can occur even with non-null causal effects [48]. Thus, one should avoid implementing NICE with overly parsimonious models. The main limitation of MSM via IPW is its lower efficiency than other methods even with stabilized weights [46••]. Though IPW using a generalized propensity score has been developed for continuous exposures, this method is less straightforward compared to binary exposures [49–51].

Besides the g-null bias, the parametric g-formula, particularly the NICE version, can require intensive computation, and typically, it is limited to marginal effects (although, in theory, effects conditional on baseline covariates could be obtained through a final model).

Discussion

Research questions about the effects of an exposure change on a health outcome are critical for clinical practice and policy making. Causal inference approaches can help articulate and estimate effects of exposure change even when the exposure is complex and incident exposure is not well-defined or captured. In this review, we discussed key considerations in defining exposure change, the total and joint effects of an exposure change, several methods to estimate total and joint effects under a target trial emulation framework and compared these methods.

Though we touched on many of the key principles, we omitted many scenarios that may be of interest to researchers. For example, we have defined the exposure change to be at one point (i.e., the visit following hypertension diagnosis). There may be value in pooling exposure change initiated at different visits after the eligibility event. In other words, an individual can contribute to multiple emulated trials at different timepoints which were discussed in the previous literature [13••, 16••, 52]. While statistically efficient [53], the pooling is only appropriate if the timing of the trial does not modify the effects [54], and clustering across individuals must be accounted for in the estimation of standard errors and confidence intervals. Multiple potential trials per person may also complicate time-dependent matching because we need to decide whether matching should be performed sequentially or simultaneously across all visits [16••, 55].

Another simplification in our example is that we dichotomize the continuous PA level to two categories: ≥ 150 min/week and < 150 min/week and ignore the fact that there is a distribution of minutes of PA within each category. To accommodate the distributions of PA when comparing two categories (called a representative intervention), participants can be randomly assigned change values from a distribution conditional on their covariates. Such methods have been

introduced for an absolute exposure level [18, 27] but have not been applied in the context of exposure change.

After applying the appropriate eligibility criteria for the exposure change definition, the sample size may become small, making it infeasible to examine the research question as currently reframed. To address this issue, one option that researchers can consider is to adapt the research question to allow the use of multiple trials per person. For example, if the eligibility event is a recent failure to control hypertension, a person may meet the eligibility criteria at multiple timepoints [53]. Another option would be to implement the IPW and parametric g-formula methods without artificial censoring, which allows borrowing information from individuals with prior non-adherence [17•]. Researchers can also choose efficient estimators (i.e., ICE, NICE rather than IPW). Lastly, study designs other than target trial emulation may be available for valid estimation of a possibly related but different causal effect that may answer a different causal question [56].

Conclusion

In settings where incident exposures may be unobservable, studying exposure changes after a meaningful eligibility event may allow researchers to use causal inference frameworks and methods to answer research questions that are relevant to clinical and policy decision making. In this review, we discussed the process of designing a study about exposure changes from defining the exposure change of interest, choosing the relevant causal effects, setting up data for the appropriate timescale, and finally, using causal inference methods to estimate causal effects. We focused on a static PA change after a hypertension diagnosis as our motivating example but also touched on study design considerations in other settings. Lastly, we described the procedures of four causal inference methods and compared their advantages and disadvantages. This review provides a practical guide that can help researchers make informed decisions when designing a study on exposure change. We hope it encourages wider utilization for causal inference methods in non-pharmacologic epidemiology.

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Declarations

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